

Diagnosis: Colorectal cancer

Treatment 1: Crizotinib

Chosen scientific papers:

Crizotinib, when combined with vemurafenib, showed rapid and marked effectiveness in treating BRAF-mutated colorectal cancer (clinical trial) [1]. Combination binimetinib/crizotinib showed poor tolerability with no objective responses observed in RASMT advanced CRC patients (clinical trial; 36 patients) [2]. Molecularly targeted treatment with crizotinib induced a rapid and sustained partial response, but disseminated tumor progression occurred after 15 months (clinical trial) [3]. The expression level of SDHB was correlated with drug sensitivity to Crizotinib (clinical trial) [4]. Crizotinib achieved a partial response (PR) with progression-free survival (PFS) of 3 months in an ALK-fusion mCRC patient (case report; 1 patient) [5]. Crizotinib delayed the progression of liver metastases and peritoneal carcinomatosis in colorectal cancer models (in vivo) [6]. Crizotinib combined with MMC showed synergistic anti-cancer effects in CRC cell lines and xenograft models (in-vivo CRC xenograft model) [7]. Crizotinib blocked the HGF/STAT3/SOX13/c-MET axis, significantly inhibiting SOX13-mediated CRC migration, invasion, and metastasis (clinical trial) [8]. Crizotinib, when combined with cetuximab or panitumumab, significantly reduces colony number and inhibits tumor growth in colorectal cancer (in vitro and in vivo) [9]. Crizotinib reduced metastasis in all models tested when used as a single agent or in combination with sunitinib (in vivo) [10].

Treatment 2: Lorlatinib

Chosen scientific papers:

Responded well to lorlatinib treatment in a case of ALK-positive colorectal cancer (clinical trial; 2 patients) [11]. Lorlatinib was effective in managing CRC patients with ALK L1196M resistance mutation, though effectiveness was not as pronounced as in NSCLC (clinical trial) [12]. An elderly patient with ALK-fusion metastatic colorectal cancer achieved a progression-free survival (PFS) of 11.5 months with lorlatinib treatment (case study; 1 patient) [13]. Lorlatinib treatment significantly suppresses polyp formation in experimental colorectal cancer by inhibiting ALKAL2/ALK signaling (in vivo) [14].

Treatment 3: Necitumumab

Chosen scientific papers:

The combination of necitumumab with mFOLFOX6 was associated with evidence of efficacy and a manageable safety profile in patients with previously untreated locally advanced or metastatic CRC (in vitro studies; 25 evaluable patients) [15].

Treatment 4: Nimotuzumab

Chosen scientific papers:

Nimotuzumab combined with chemotherapy significantly improves the disease control rate and survival benefit in patients with solid tumors (retrospective clinical trial) [16]. The efficacy of this regimen (pCR = 19.0%) was significantly higher (Prospective phase II trial; 21 patients) [17]. The efficacy of nimotuzumab combined with CT in the treatment of

advanced colorectal cancer was slightly better than that of cetuximab, and with mild adverse reactions (retrospective study; 205 patients) [18]. Nimotuzumab showed less effectiveness compared to [225Ac]Ac-macropa-nimotuzumab in KRAS mutant and BRAFV600E mutant CRC models (in vitro; in vivo) [19]. Nimotuzumab had a minimal effect on tumor growth but no complete cure, while nimotuzumab-PEG6-DM1 showed more tumor necrosis and complete tumor regression in some cases (in vitro) [20]. Nimotuzumab-PEG6-DM1-High resulted in 2/5 complete cure in colorectal cancer model (in vitro and in vivo; 5 patients) [21]. Nimotuzumab-PEG6-DM1 treatment showed complete remission in 1/5 mice and prevented metastatic spread in 4/5 mice (in vivo) [22]. Nimotuzumab is one of the drugs studied in colorectal cancer showing promising results (clinical trial) [23]. Nimotuzumab is one of the anti-EGFR antibodies under clinical development for the treatment of metastatic colorectal cancer (clinical trial) [24].

Treatment 5: Panitumumab

Chosen scientific papers:

Sotorasib 960 mg-panitumumab showed improved outcomes and is considered a new standard-of-care treatment for KRAS G12C-mutated chemorefractory mCRC (clinical trial; 160 patients) [25]. Sotorasib combined with panitumumab significantly prolonged progression-free survival in treated CRC patients (clinical trial) [26]. Panitumumab combined with chemotherapy has demonstrated significant improvements in response rates and overall survival (clinical trial) [27]. Panitumumab/FOLFOX treatment in RAS wild-type mCRC showed ctDNA clearance in 80.2% of patients with baseline detectable ctDNA, associated with better response depth (clinical trial; 154 patients) [28]. The modified mCCS supports the effectiveness of first-line therapy with panitumumab in combination with FOLFOX/FOLFIRI for RAS-WT mCRC (clinical trial; 646 patients) [29]. Panitumumab, when combined with other drugs like Bevacizumab or used with Irinotecan, improved survival outcomes in colorectal cancer patients (randomized controlled trial; 5,911 patients) [30]. Panitumumab with chemotherapy may benefit KRAS wild-type metastatic colorectal cancer patients, but not all respond to this therapy (clinical trial) [31]. Panitumumab showed an ORR of 10.9% (95% CI, 0.00-26.82%) and a DCR of 63.05% (95% CI, 52.13-73.97%) (clinical trial) [32]. Patients with KRAS wild-type metastatic colorectal cancer treated with panitumumab have improved progression-free survival and overall survival compared to best supportive care (clinical trial; 21 patients) [33]. FOLFOX/FOLFIRI-panitumumab increased response at the cost of more toxic effects in RAS/BRAFV600E wild-type, left-sided tumors (clinical trial; 118 patients) [34].

Treatment 6: Cetuximab

Chosen scientific papers:

Triple therapy with cetuximab offers meaningful improvements in survival and tumor response in BRAF V600E-mutated CRC (clinical trial; 487 patients) [35]. Cetuximab, when combined with encorafenib and nivolumab, showed an overall response rate of 50% in patients with microsatellite stable BRAFV600E metastatic colorectal cancer (clinical trial; 26 patients) [36]. Cetuximab combined with mFOLFOX6 showed an objective response rate (ORR) of 77.5% in RAS wild-type metastatic colorectal cancer (clinical trial; 347 patients) [37]. The combination of encorafenib and cetuximab offers significant clinical benefits, improving PFS and OS, while providing manageable safety and important QoL advantages (Clinical trial; 97 patients) [38]. Cetuximab plus FOLFIRI may confer a survival advantage over bevacizumab in KRAS wild-type mCRC without PTR, including during later-line therapy (clinical trial; 13,473 patients) [39]. Cetuximab in combination with encorafenib improved

overall survival, objective response rate, and progression-free survival in previously treated BRAFV600E-mutant metastatic colorectal cancer (clinical trial; 220 patients) [40]. Cetuximab combined with encorafenib and chemotherapy showed significant improvements in overall survival, progression-free survival, and objective response rate in patients with BRAF-mutated colorectal cancer (clinical trial; 24,327 patients) [41]. Cetuximab showed durable treatment benefit when used as first-line therapy for left-sided, RAS wild-type metastatic colorectal cancer, especially after switching due to skin toxicity (Clinical trial; 210 patients) [42]. Cetuximab, in combination with encorafenib and mFOLFOX6, showed improved progression-free survival and overall survival in patients with BRAF V600E-mutant metastatic colorectal cancer (clinical trial) [43]. Pathological complete response (pCR) was achieved after treatment with mFOLFOX6 plus cetuximab in a case of synchronous peritoneal metastasis from colorectal cancer (clinical trial) [44].

Treatment 7: Brigatinib

Chosen scientific papers:

Brigatinib demonstrates a considerable anti-cancer effect in CRC cells in vitro and in vivo (in vitro and in vivo) [45].

Treatment 8: Ramucirumab

Chosen scientific papers:

Ramucirumab prolongs overall survival and progression-free survival compared with FOLFIRI alone in patients with metastatic colorectal cancer (clinical trial) [46]. Ramucirumab showed a hazard ratio of 0.9206 (95% CI 0.6504-1.303) for progression-free survival in metastatic colorectal cancer (clinical trial; 6805 patients) [47]. Ramucirumab plus FOLFIRI therapy is the standard second-line treatment for metastatic colorectal cancer (clinical trial; 147 patients) [48]. FOLFIRI plus ramucirumab significantly improved OS in patients with metastatic colorectal carcinoma refractory to fluoropyrimidine, OHP, and bevacizumab (clinical trial; 125 patients) [49]. Patients treated with ramucirumab/FOLFIRI demonstrated significant improvement in PFS (5.7 months) versus placebo/FOLFIRI (4.5 months) (clinical trial; 410 patients) [50]. FOLFIRI plus RAM showed clinical efficacy and safety comparable to FOLFOXIRI plus RAM in untreated mCRC (Randomized phase II clinical trial; 122 patients) [51]. The addition of ramucirumab to irinotecan and cetuximab improved PFS and disease control rate, showing the combination is feasible and effective (randomized phase II study; 102 patients) [52]. Ramucirumab could be more effective than bevacizumab in patients with high VEGF-D (clinical trial; 345 patients) [53]. Ramucirumab, when combined with FOLFIRI, showed improved overall survival (OS) and progression-free survival (PFS) in patients with metastatic colorectal cancer (mCRC) (clinical trial; 410 patients) [54]. Ramucirumab combined with FOLFIRI regimen was studied in metastatic colorectal cancer patients (clinical trial; 1072 patients) [55].

Treatment 9: Pomalidomide

Chosen scientific papers:

Pomalidomide was used in combination with deltatizonone to design PROTAC PDE1 α degraders, which showed improved antiproliferative potency in KRAS mutant cells (in vitro and in vivo) [56]. Pomalidomide significantly inhibits the metastatic capability of colorectal carcinoma cells (in vitro and in vivo) [57].

Treatment 10: Toremifene

Chosen scientific papers:

Toremifene was a potent inhibitor of AC activity (in vitro) [58].

Treatment 11: Denosumab

Chosen scientific papers:

Treatment 12: Duvelisib

Chosen scientific papers:

Duvelisib synergized with RT to suppress tumor growth and induce durable immune memory in the CT26 model (in vivo) [59].

Treatment 13: Inavolisib

Chosen scientific papers:

Inavolisib shows increased efficacy in colorectal cancer tumor xenograft models, particularly when combined with cobimetinib (in vitro and in vivo) [60].

Treatment 14: Bevacizumab

Chosen scientific papers:

FPSIR predicts clinical therapeutic responses and survival outcomes in patients with metastatic colorectal cancer undergoing first-line bevacizumab-containing chemotherapy (clinical trial; 364 patients) [61]. Bevacizumab showed a moderately improved OS (HR 0.71, 95% CI, 0.42-1.2) and significantly improved PFS (HR 0.49, 95% CI, 0.29-0.85) in metastatic colorectal cancer (clinical trial; 70 patients) [62]. FTD-TPI combined with bevacizumab showed greater OS benefit of 17.04 months in third-line mCRC (clinical trial; 3,978 patients) [63]. Bevacizumab, as part of the EZFB regimen, significantly improved progression-free survival and overall survival in patients with third-line mCRC (clinical trial; 112 patients) [64]. Bevacizumab combined with chemotherapy showed reduced progression free survival in BRAFV600E patients, indicating limited effectiveness (clinical trial; 146 patients) [65]. FTD-TPI+bev improved overall survival in patients with metastatic colorectal cancer compared with FTD-TPI monotherapy (clinical trial; 529 patients) [66]. Bevacizumab combined with chemotherapy in colorectal cancer patients with brain metastases achieved a median OS of 20.6 months (clinical trial) [67]. Bevacizumab combined with two-drug chemotherapy significantly prolonged progression-free survival (PFS) in patients with RAS-mutant mCRC (clinical trial; 275 patients) [68]. Bevacizumab plus FOLFIRI was associated with better survival outcomes, particularly in patients who underwent PTR, likely due to reduced tumor vascular burden (clinical trial; 13,473 patients) [69]. The combination of Nivolumab with FOLFOXIRI/Bevacizumab showed an objective response rate of 76.7% and a disease control rate of 97.3% in advanced RAS/BRAF-mutated colorectal cancer (clinical trial; 73 patients) [70].

Treatment 15: Anastrozole

Chosen scientific papers:

Treatment 16: Letrozole

Chosen scientific papers:

Treatment with letrozole resulted in endoscopic and symptomatic improvement in her colorectal stricture (case report; 1 patient) [71]. Letrozole may influence colon cancer progression by reducing estrogen synthesis and modulating immune-related pathways such as GPR30-Akt signaling and PD-L1 expression (in vitro, in vivo) [72].

Treatment 17: Tamoxifen

Chosen scientific papers:

Tamoxifen is associated with an increased risk of colorectal cancer in postmenopausal women with breast cancer (clinical trial; 8893 patients) [73]. Tamoxifen clearly showed a reduction in cancer incidence, but it was not widely used due to a lack of evidence for an overall long-term benefit (clinical trial) [74]. Tamoxifen induces tumor formation in the KPC: APC model, mimicking KRAS mutated colorectal cancer (in vivo) [75]. TAM effectively inhibited the viability and migration ability of DLD-1 colorectal cells and promoted apoptosis (in vitro) [76]. Tamoxifen reduces the outgrowth of colon cancer liver metastasis and increases liver cytotoxic T cell accumulation (in vitro, in vivo) [77]. Tamoxifen treatment in KPC:APC mice caused extensive inflammation in the colon and led to a rapid decline in health, with rectal bleeding and cachexia observed (in vivo) [78]. Tamoxifen suppresses cancer stemness in CRC by downregulating LNX1 and reducing colonosphere formation (in vivo) [79]. Tamoxifen may negatively impact CRC patient outcomes by increasing STS activity and local estrogen availability (in vitro) [80]. Tamoxifen-induced Cre expression and inflammation dynamics in the colon were assessed, showing its role in promoting tumor development when combined with mild inflammation (in vivo) [81]. Tamoxifen was used 5 times during the first DSS treatment window in the AOM/DSS treatment regime (in vivo) [82].

Treatment 18: Fulvestrant

Chosen scientific papers:

Fulvestrant highly enhanced Il-6 expression in C26 cells, suggesting an anti-inflammatory effect (in vitro and in vivo) [83]. Fulvestrant significantly increased STS activity in HCT116 cells when co-administered with E2 (in vitro) [84]. Fulvestrant showed a significant and dose-dependent reduction in the number of colorectal cancer liver metastases and enhanced the therapeutic effect of anti-PD1 immunotherapy (in vitro and in vivo) [85]. Fulvestrant is mentioned as a literature-supported cancer drug for colorectal cancer (review) [86].

Treatment 19: Exemestane

Chosen scientific papers:

Treatment 20: Buserelin

Chosen scientific papers:

Buserelin did not induce a significant stimulatory influence on the HI variants of the LoVo and HCT-15 cells, but showed a marked tropic response in cells cultured in E2 and/or gastrin + 1% FCS-supplemented medium (in vitro) [87].

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